

Active Collaborative Learning - 1
Networks and System Security - Autumn, 2026

Date: 24.08.2026

Max. marks: 50

Duration: 1.5 hours

Answer all of the following questions:

1. Alice and Bob are engineers at a remote weather-monitoring station. Alice's sensor node continuously transmits climate data packets to Bob's base station over a shared wireless channel that is also used by two other sensor nodes (Carol and Dave) in the same range. The link is noisy, some packets get completely lost during heavy rain, while others arrive with corrupted bits due to lightning-induced interference. Bob's receiver has a limited buffer and cannot process data arbitrarily fast. Each device on the network has been assigned a unique hardware identifier by its manufacturer.

Answer the following:

- (a) Alice's node sends data faster than Bob's buffer can consume it, occasionally causing overflow. Suggest a suitable flow control mechanism to solve this, and explain how it prevents buffer overflow. Compare it briefly with an alternative flow control scheme in terms of efficiency.
- (b) During a storm, some packets from Alice never reach Bob at all, while others arrive with flipped bits. Identify and explain two different mechanisms — one to detect a completely lost packet, and one to detect bit-level corruption within a received packet.
- (c) Since Alice, Carol, and Dave all share the same wireless channel, their transmissions sometimes collide. Name and explain the access control protocol that would let them share the channel without simultaneous transmission, and describe what happens when a collision is detected.
- (d) Bob wants to identify each incoming packet's source node uniquely at the hardware level, even before any IP addressing is considered. Explain the structure of the identifier used for this purpose, including how uniqueness is guaranteed globally.

(3+2+2+3)

2. A network engineer is troubleshooting communication between Host A and Host B. The exact path between them is not known. During testing, the engineer observes the following:

Observation 1: A packet sent from A with TTL = 3 produces: ICMP TTL Exceeded

Observation 2: The same destination is tested again with TTL = 5. This time, the sender receives an : ICMP Echo Reply

Observation 3: A third packet sent toward the same destination produces: ICMP Destination Unreachable

The engineer asks five students for their conclusions:

Student 1: "There must be exactly three routers between A and B."

Student 2: "The Echo Reply proves that every packet from A can reach B."

Student 3: "TTL Exceeded and Destination Unreachable mean the same thing because both indicate packet failure."

Student 4: "If a packet enters a routing loop, TTL can prevent it from circulating forever."

Student 5: "If ping succeeds, there cannot be a Network Layer problem."

Answer the following:

(a) Analyze Student 1's conclusion. Is it possible to determine the exact number of routers between A and B from Observation 1 alone? Explain what can actually be inferred.

(b) Analyze Student 2 and Student 5 together. What does an ICMP Echo Reply actually establish? Give one important conclusion that can be made and one conclusion that cannot be made.

(c) Student 3 claims that TTL Exceeded and Destination Unreachable mean the same thing. Explain why the Network Layer uses different feedback messages for these situations.

(d) Now assume the actual forwarding path contains a loop:

$R1 \rightarrow R2 \rightarrow R3 \rightarrow R1 \rightarrow R2 \rightarrow R3 \rightarrow \dots$

Explain what would happen to a packet travelling through this loop and how TTL changes the situation.

(e) Based only on the three observations, identify one conclusion that the engineer can confidently make and one conclusion that would be an unjustified assumption. Explain your reasoning. (2+2+2+2+2)

3. The network G has been allotted the classless block 192.168.40.0/24. The network administrator wants to divide this block into 4 equal-sized subnets to accommodate four departments — CSE, ECE, ME, and Admin.

(a) Determine the subnet mask used for this division. For each of the 4 subnets, clearly state the network id, the usable host range, and the directed broadcast id.

(b) Host A has been assigned the IP address 192.168.40.75. Identify which of the 4 subnets Host A belongs to.

(c) A junior network engineer proposes allotting the IP address block 192.168.40.16 – 192.168.40.29 (14 addresses) to a fifth, smaller department. Using the rules governing a valid CIDR block, determine whether this proposal is valid. Justify your answer with respect to each applicable rule.

(d) For each of the following network operations, identify the exact OSI layer responsible for performing it, and justify your answer in one line.

(i) Compressing outgoing data (zipping it) and translating it into a common character set before transmission.

(ii) Splitting a large message into smaller segments at the sender, and demultiplexing an incoming message to the correct destination process among several running processes at the receiver.

(iii) A router examining the destination network id of an incoming packet to decide which outgoing link to forward it on.

(iv) Converting a stream of bits (1s and 0s) into electrical voltage levels for transmission over a copper cable.

(e) Suppose Host A now uses a browser to visit a website hosted on a server in another network, over HTTP. Answer the following short questions.

(i) State the port number used by the DNS protocol, and the port number used by the HTTP protocol.

(ii) Is HTTP a stateful or a stateless protocol? Explain in 1–2 lines what this implies for how the server treats client requests.

(iii) Distinguish between HTTP 1.0 and HTTP 1.1 in terms of the type of connection each uses.

(2+2+2+2+2)

4. A client wants to establish a TCP connection with a server and then transfer a file. The following events are observed:

Client → Server: SYN, Seq = 5000

Server → Client: SYN + ACK, Seq = 9000

Client → Server: ACK

After the connection is established, the client sends a TCP segment containing 1000 bytes of data. The next segment contains another 800 bytes, but this segment is lost in the network. The following segment containing 600 bytes reaches the server successfully. The server then sends some data of its own while also acknowledging data received from the client. The client later finishes sending its data and initiates connection termination.

Answer the following:

- (a) Determine the sequence number and acknowledgement number of the client's final ACK in the three-way handshake. Explain why the SYN affects the sequence number even though it carries no application data.
- (b) The client's first data segment contains 1000 bytes. If its sequence number is 5001, what acknowledgement number should the server generate after receiving it correctly? Explain the relationship between sequence number, data length and acknowledgement number.
- (c) The second data segment is lost, but the following 600-byte segment arrives at the server. What acknowledgement should the server send? Why does the server not simply acknowledge the data that arrived later?
- (d) The server has data of its own to send while acknowledging the client's data. Explain how piggybacking can be used in this situation and why it can be more efficient than sending a separate acknowledgement packet.
- (e) Suppose the receiver is temporarily unable to process data as quickly as the sender is transmitting it. Explain how TCP flow control/window size helps prevent the sender from overwhelming the receiver.
- (f) Finally, the client wants to terminate its connection. Explain the role of the FIN flag and whether FIN consumes a sequence number. (2+2+2+1+2+1)

5. Answer the following.

- (a) For the IP address '137.156.31.68/19' determine the following: (i) the subnet address, (ii) directed broadcast address for the subnet, (iii) number of possible hosts on that subnet, (iv) maximum number of subnets.
- (b) A large number of consecutive IP addresses are available starting at 54.128.128.0. Suppose that five organizations, A, B, C, D and E, request 1000, 3000, 500, 16000 and 8000 addresses, respectively, and in that order. For each of these, give the first IP address assigned, the last IP address assigned, and the mask in the w.x.y.z/s notation.
- (c) Consider a host with IP address 10.10.10.10 in a network 10.10.10.0/24 with gateway router 10.10.10.1. Where will the host forward packets if the destinations are: (i) 10.10.10.11, (ii) 10.10.11.10, and (iii) 10.11.10.10? (2+5+3)

